

AZURE MANAGEMENT GROUP HIERARCHY DESIGN

Team Details:

Team Number: T257

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1. Problem Statement

Organizations using Microsoft Azure often manage multiple departments, subscriptions, and cloud resources. Without a properly planned management structure, it becomes difficult to maintain consistent governance, security, access control, and compliance across different departments.

The project aims to design an Azure Management Group hierarchy for four departments and demonstrate how governance policies can be applied and inherited throughout the hierarchy. The proposed hierarchy will organize departments and their subscriptions in a structured manner, allowing common policies to be applied at a higher level and inherited by the departments below it.

The project will also analyze two major challenges associated with Azure management group design. First, changing the hierarchy after subscriptions and policy assignments have already been configured can be disruptive. Second, policies inherited from parent management groups may combine with policies assigned at lower levels, resulting in unexpected effective policy outcomes.

Therefore, the project focuses on designing a scalable and manageable Azure hierarchy and applying policies correctly to ensure centralized governance while allowing department-specific requirements.

2. Objectives

The main objectives of the project are:

1. To design an Azure Management Group hierarchy for four different departments.
2. To organize Azure subscriptions under appropriate management groups.
3. To establish a clear parent-child relationship between the management groups.
4. To apply common governance policies at the higher management group level.
5. To demonstrate policy inheritance from parent management groups to child management groups and subscriptions.
6. To apply department-specific policies where required.
7. To understand how multiple inherited and local policies affect the effective policy applied to a subscription or resource.
8. To identify the impact of modifying the management group hierarchy after subscriptions and policy assignments have been created.
9. To provide a structured and scalable governance model for managing Azure resources.
10. To improve consistency, security, and centralized control across different departments.

3. Proposed System

The proposed system uses Azure Management Groups to organize the cloud environment into four departments.

- Human Resources (HR)

- Finance
- Information Technology (IT)
- Sales

Subscriptions are organized under their respective departments.

The proposed structure is:

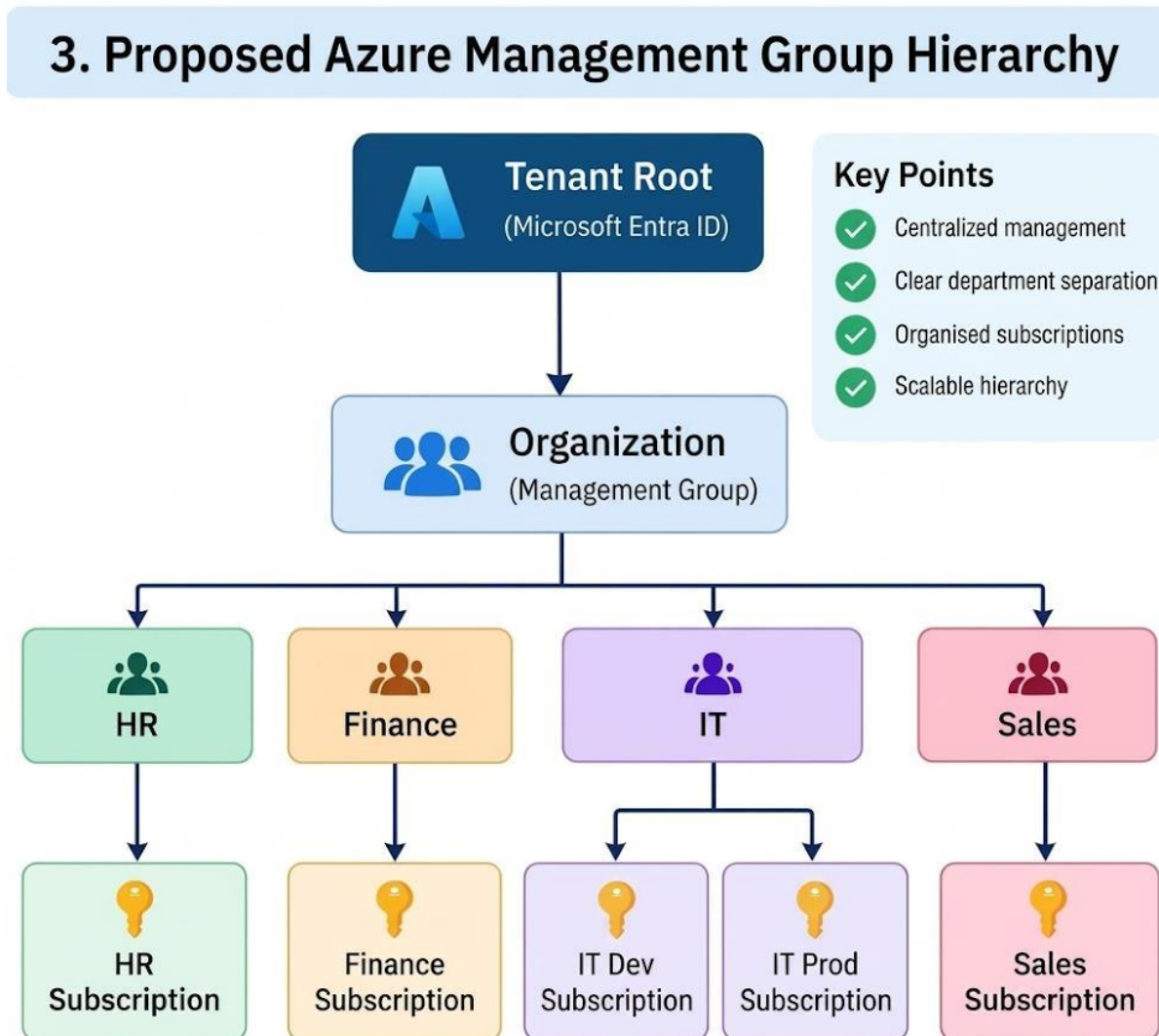


Figure 1: Proposed Azure Management Group Hierarchy

The Organization management group acts as the parent management group. HR, Finance, IT, and Sales are child management groups. Subscriptions are placed under the appropriate department according to their purpose.

Policies can be assigned at the Organization level so that common governance requirements are inherited by all departments. Additional policies can be assigned at individual department levels when specific requirements exist.

4. Project Methodology

The project follows a structured methodology to design and evaluate the Azure Management Group hierarchy and its governance policies.

4.1 Requirement Analysis

The first step is to identify the departments, subscriptions, governance requirements, and policy requirements that need to be managed. The project considers four departments: HR, Finance, IT, and Sales.

4.2 Hierarchy Design

A parent Organization Management Group is designed with four child management groups representing the departments. The hierarchy establishes a clear parent-child relationship between the organization, departments, and subscriptions.

4.3 Subscription Organization

Subscriptions are organized under their respective departmental management groups. For example:

- HR Subscription → HR Management Group
- Finance Subscription → Finance Management Group
- IT Development Subscription → IT Management Group
- IT Production Subscription → IT Management Group
- Sales Subscription → Sales Management Group

4.4 Policy Assignment

Common governance policies are assigned at the Organization management group level. Examples include:

- Restricting permitted Azure regions
- Requiring resource tags
- Enforcing approved resource configurations
- Maintaining organizational governance standards

4.5 Department-Specific Policy Assignment

Additional policies are assigned to individual departments when specific requirements exist. For example, the IT department can have an additional policy for resource tagging or configuration. Therefore, IT can have both organization-level policies and IT-specific policies.

4.6 Policy Inheritance Testing

The project verifies whether policies assigned at the parent management group are inherited by the applicable child management groups and subscriptions, following the flow: Organization Policy → HR / Finance / IT / Sales → Department Subscriptions.

4.7 Effective Policy Analysis

The project analyzes how inherited policies and department-specific policies combine to determine the effective governance rules for a subscription or resource. For example, an IT subscription may be affected by the Organization-level location policy and the IT-specific resource tagging policy.

4.8 Bottleneck Analysis

The project analyzes two major bottlenecks: the impact of changing the hierarchy after subscriptions and policies have already been assigned, and unexpected results caused by the combination of inherited and department-specific policies.

Methodology Flow

Requirement Analysis → Hierarchy Design → Subscription Organization → Policy Assignment → Policy Inheritance Testing → Effective Policy Analysis → Bottleneck Analysis → Final Governance Model

This methodology is consistent with the hierarchy, policy, effective-policy, and hierarchy-change activities specified in the project requirements and use cases.

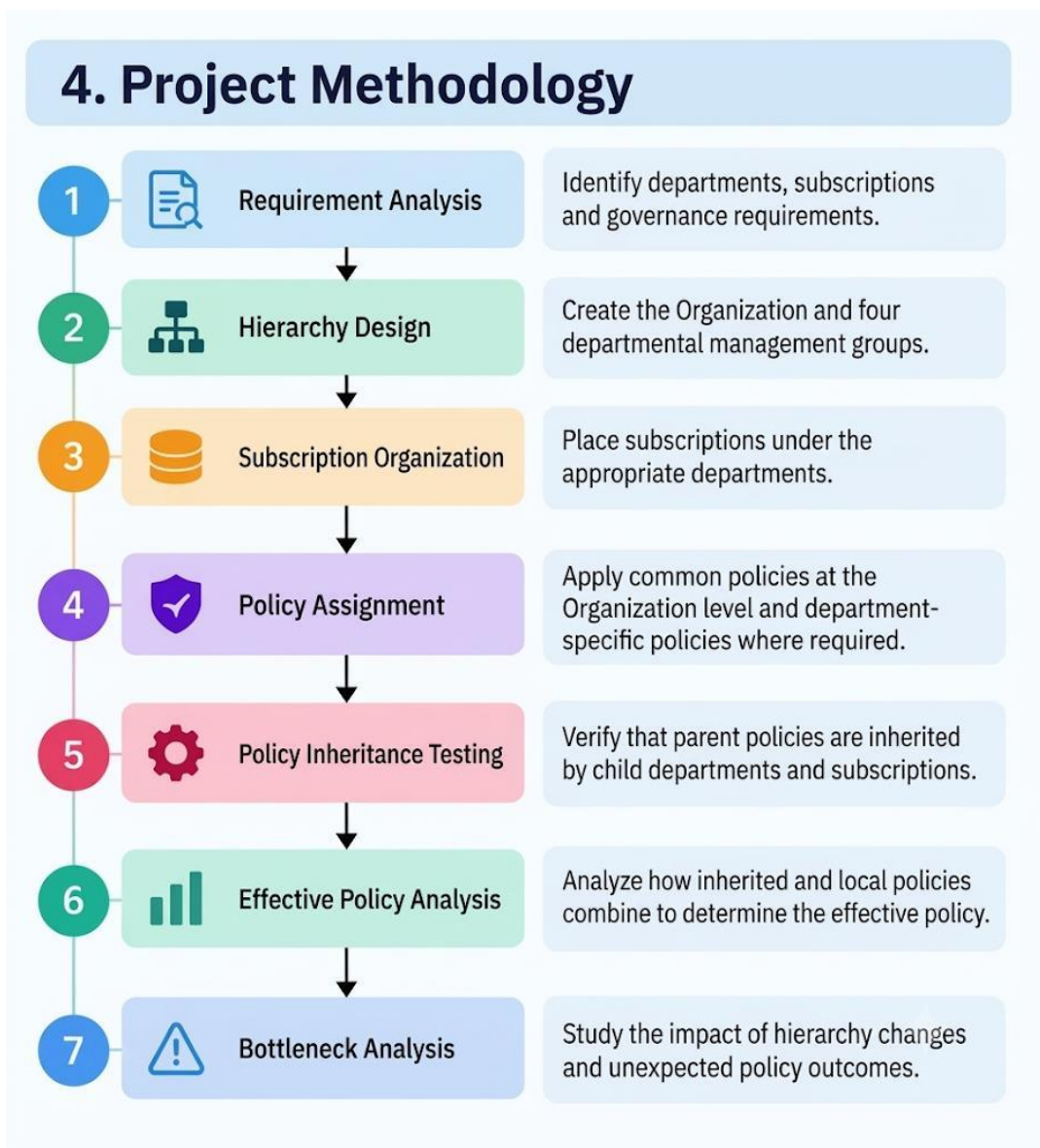


Figure 2: Project Methodology Flow

5. Requirements

5.1 Functional Requirements

FR1 – Management Group Creation

The system shall provide a hierarchy containing a parent management group and four child management groups representing four departments. The four departments considered in the proposed design are:

- Human Resources (HR)
- Finance
- Information Technology (IT)
- Sales

FR2 – Subscription Organization

The system shall allow subscriptions to be organized under the appropriate departmental management groups. Examples:

- HR Subscription → HR Management Group
- Finance Subscription → Finance Management Group
- IT Development Subscription → IT Management Group
- IT Production Subscription → IT Management Group
- Sales Subscription → Sales Management Group

FR3 – Hierarchy Management

The system shall maintain a clear parent-child relationship between management groups, subscriptions, and resources. This relationship should allow administrators to clearly identify which department a subscription belongs to and how resources are organized within the Azure environment.

FR4 – Policy Assignment

The system shall support the assignment of governance policies at the Organization or management group level. Examples of common policies include:

- Restricting permitted Azure regions
- Requiring resource tags
- Enforcing approved resource configurations
- Maintaining organizational governance standards

FR5 – Policy Inheritance

Policies assigned at a parent management group shall be inherited by applicable child management groups and subscriptions, following the flow: Organization Policy → HR | Finance | IT | Sales → Department Subscriptions. This allows common governance rules to be applied centrally throughout the hierarchy.

FR6 – Department-Specific Policies

The system shall allow additional policies to be assigned at the department level, following the flow: Organization → IT → IT-Specific Policy. The IT department will therefore be affected by both the inherited organization-level policy and its own department-level policy.

FR7 – Effective Policy Evaluation

The system shall allow the project team to analyze how inherited and department-specific policies combine and determine the effective governance rules for a subscription or resource. For example, an IT subscription may be subject to both the inherited Organization-level policy and the IT-specific policy. The resulting combination represents the effective governance rules applicable to the IT subscription.

FR8 – Hierarchy Change Analysis

The system shall analyze the impact of moving a subscription or changing its position in the management group hierarchy after policies and assignments have been configured. The analysis should identify how such

a hierarchy change can affect the policies, governance structure, and assignments applicable to the subscription.

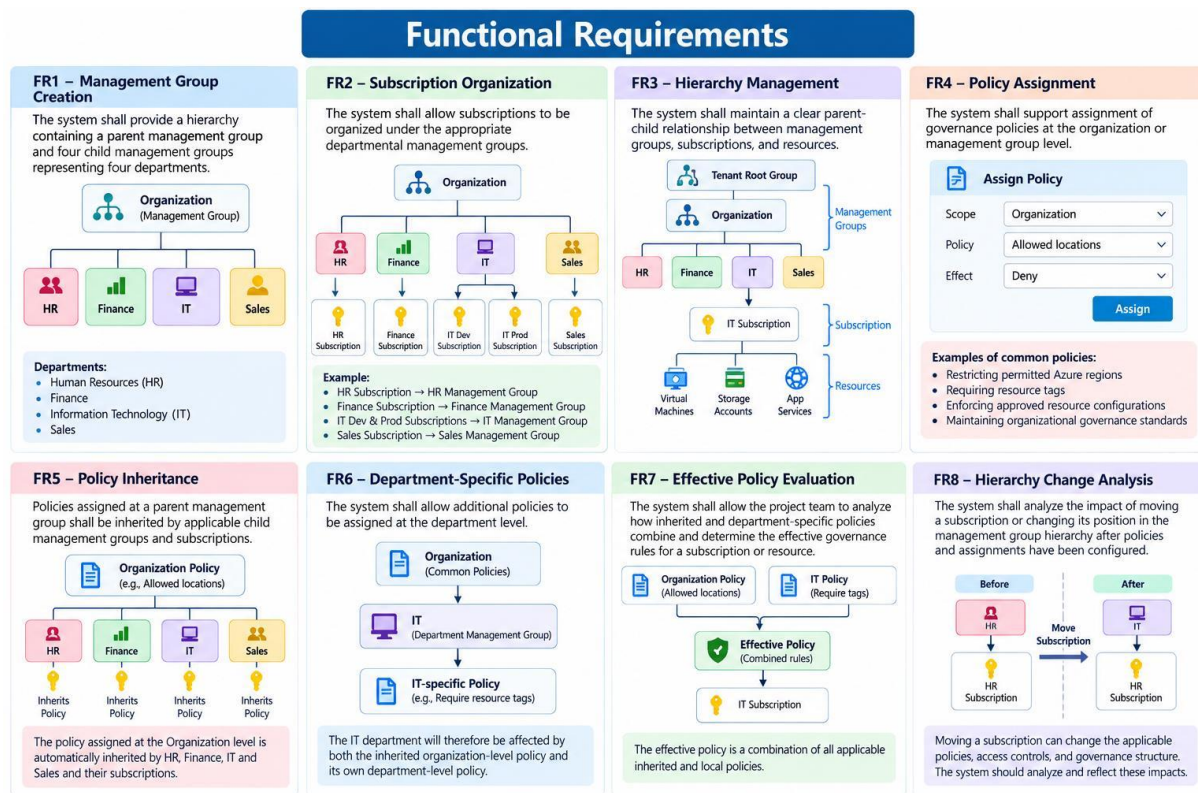


Figure 3: Functional Requirements Overview

5.2 Non-Functional Requirements

NFR1 – Scalability

The hierarchy should be designed so that additional departments, subscriptions, and resources can be added in the future without requiring a complete redesign of the existing structure.

NFR2 – Security

The design should support centralized governance and help ensure that organizational policies are consistently applied across the Azure environment.

NFR3 – Manageability

The hierarchy should be simple and well organized so that administrators can easily understand and manage departments, subscriptions, and their relationships.

NFR4 – Reliability

Policies should be consistently applied according to their assigned scope and inheritance structure. The system should provide predictable governance behavior across the hierarchy.

NFR5 – Maintainability

The hierarchy should be designed carefully so that future changes do not unnecessarily disrupt existing subscriptions, assignments, and governance policies.

NFR6 – Compliance

The system should support enforcement of organizational standards through Azure Policy and help ensure that resources follow the required governance rules.

6. Use Cases

Use Case 1 – Design Hierarchy for Four Departments

Actor: Azure Administrator

Description: The administrator creates a management group structure for four departments and organizes their subscriptions under the appropriate department.

Process:

11. Create the parent Organization management group.
12. Create four child management groups.
13. Assign the appropriate subscriptions to each department.
14. Verify the parent-child relationships.
15. Ensure that the hierarchy is clearly organized.

Expected Result: A structured Azure Management Group hierarchy is created for all four departments.

Use Case 2 – Apply Policy Inheritance

Actor: Azure Administrator

Description: The administrator applies common governance policies at the parent management group level and verifies that they are inherited by child management groups and subscriptions.

Process:

16. Create or select an appropriate Azure Policy.
17. Assign the policy to the Organization management group.
18. Allow the policy to apply to child management groups.
19. Verify policy inheritance for HR, Finance, IT, and Sales.
20. Apply additional policies to selected departments if required.
21. Analyze the resulting effective policies.

Expected Result: Common governance policies are consistently inherited throughout the management group hierarchy.

7. Key Bottlenecks / Challenges

7.1 Hierarchy Changes After Assignments

Changing the management group hierarchy after subscriptions, policies, or access assignments have already been configured can be disruptive. For example, moving a subscription from the HR management group to the IT management group can cause it to become subject to the governance policies and access structure associated with IT.

Therefore, the hierarchy should be carefully planned before assigning subscriptions and policies.

7.2 Unexpected Policy Inheritance

Policies assigned at a parent management group can be inherited by child management groups and subscriptions. For example:

- Organization Policy: Resources must use approved locations.
- IT Policy: Resources must contain specific tags.

An IT subscription may therefore be subject to both policies. This can result in unexpected effective policy results if administrators do not understand the inheritance structure.

Therefore, policy assignments and their inheritance should be reviewed carefully before deployment.

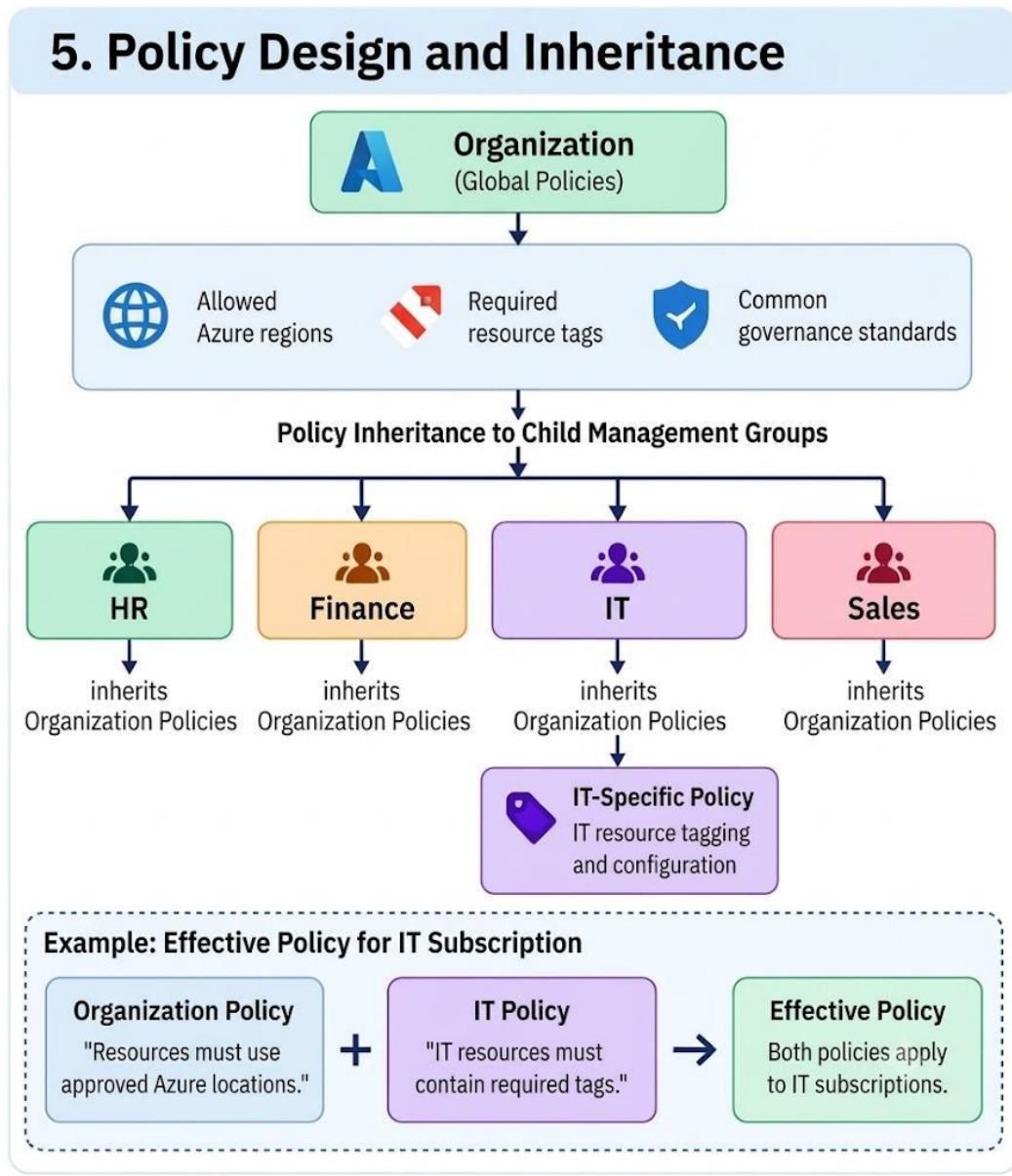


Figure 4: Policy Design and Inheritance

8. Expected Outcome

The expected outcome of the project is a well-structured Azure Management Group hierarchy containing four departments with appropriately organized subscriptions.

The project will demonstrate:

- Centralized Azure governance.
- Proper organization of departments and subscriptions.
- Parent-child management group relationships.
- Application of organization-level policies.
- Correct inheritance of policies.

- Application of department-specific policies.
- Analysis of effective policy results.
- Understanding of the impact of hierarchy changes.
- A scalable structure for future departments and subscriptions.

The proposed design will help organizations manage Azure resources in a structured, secure, and maintainable manner.

9. Conclusion

Azure Management Groups provide an effective way to organize multiple subscriptions and apply governance policies at scale. By designing a hierarchy for HR, Finance, IT, and Sales, common organizational policies can be managed centrally while allowing departments to have additional requirements.

The project focuses on correct hierarchy design and policy inheritance. It also considers important challenges such as disruptive hierarchy changes and unexpected effective policy results caused by inherited policies. A carefully planned management group hierarchy can therefore improve governance, consistency, security, and manageability of Azure resources.

10. Prototype

The following prototype screens illustrate the Azure Management Group hierarchy, policy assignment workflow, and compliance reporting as implemented in the Azure Portal.

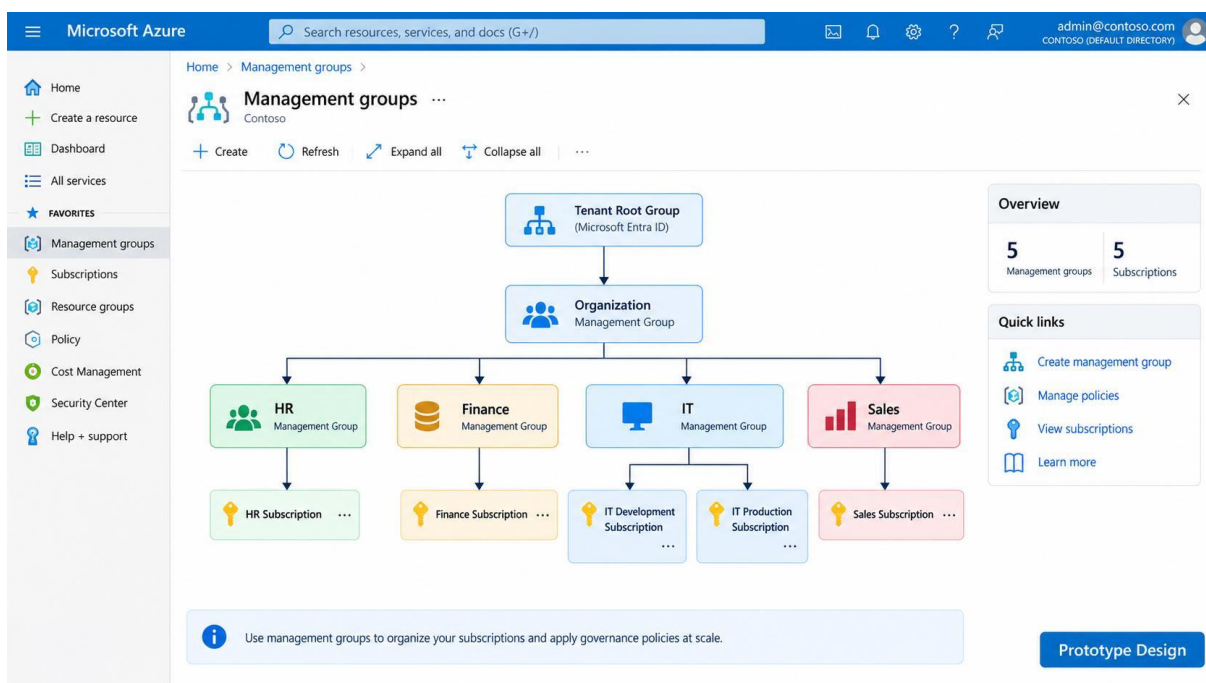


Figure 5: Management Groups Overview

Microsoft Azure Search resources, services, and docs (G+/)

admin@contoso.com
CONTOSO (DEFAULT DIRECTORY)

Home > Policy > Assignments >

Create policy assignment

Basics Parameters Remediation Non-compliance messages Review + create

Scope
Select the management group, subscription or resource group where this policy will be assigned.

Scope * ...
[Learn more about setting the scope](#)

Policy definition
Choose the policy definition you want to assign.

Policy definition * ...
[View policy definition](#)

Assignment details

Assignment name *

Description

Enforcement mode ☒ Default ☐ Do not enforce 49/500

Assignment identity

[Review + create](#) [< Previous](#) [Next: Parameters >](#)

Policy definition details

Allowed locations
Built-in

Description
Restricts the locations where resources can be deployed. Helps ensure compliance with organizational requirements.

Category
Locations

Version
1.0.0

Related information

[View definition in library](#)
[Learn more about this policy](#)
[View compliance results](#)

This is a prototype interface for illustration purposes.

[Prototype Design](#)

Figure 6: Create Policy Assignment

Microsoft Azure Search resources, services, and docs (G+/)

admin@contoso.com
CONTOSO (DEFAULT DIRECTORY)

Home > Management groups > IT

IT | Subscriptions

Management group

[Add subscription](#) [Refresh](#) [Export](#)

Search subscriptions Showing 1 to 2 of 2 subscriptions

State: All Subscription type: All Location: All

Name	Subscription ID	Subscription type	State	Location	...
IT Development	a1b2c3d4-e5f6-7890-1234	Azure Plan	Active	East US	...
IT Production	b2c3d4e5-f6a7-8901-2345	Azure Plan	Active	East US	...

IT
Management group

Overview

2	0	15
Subscriptions	Resource groups	Resources

Management group details

Name IT

ID /providers/Microsoft.Management/managementGroups/IT

Parent Organization

Type Management group

Quick links

[Add subscription](#)
[View policies](#)
[View role assignments](#)
[Download report](#)

Use management groups to organize subscriptions and apply governance at scale.

[Prototype Design](#)

Figure 7: IT Management Group – Subscriptions

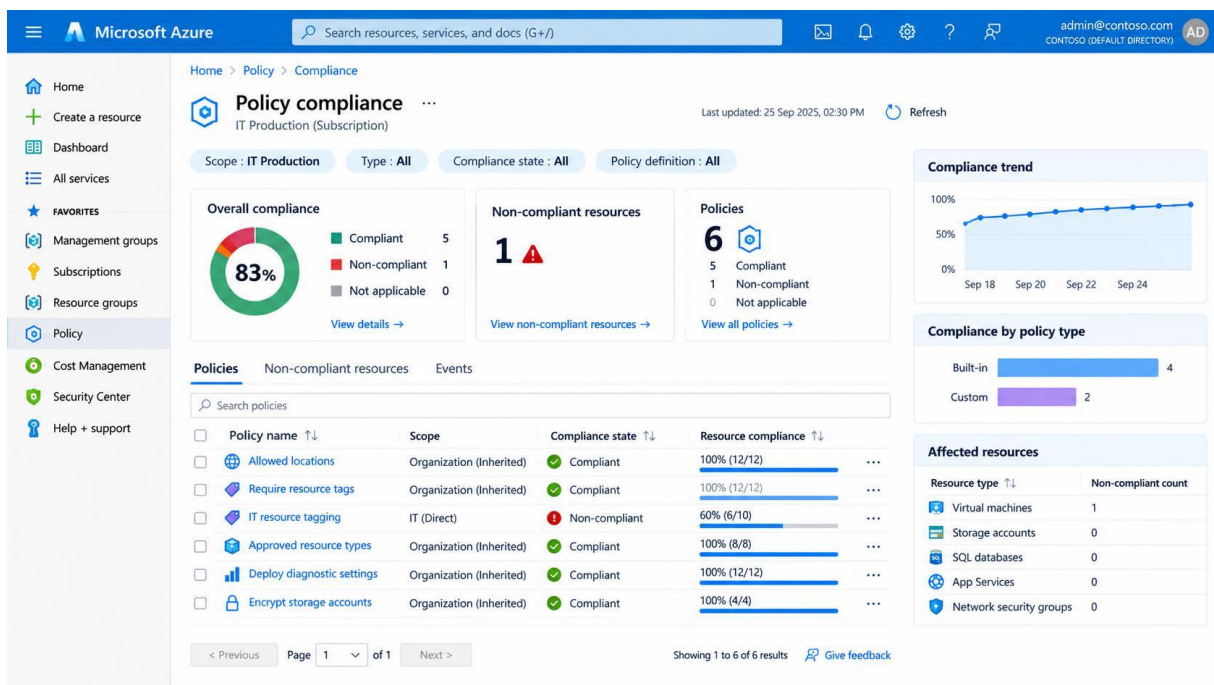


Figure 8: Policy Compliance Dashboard

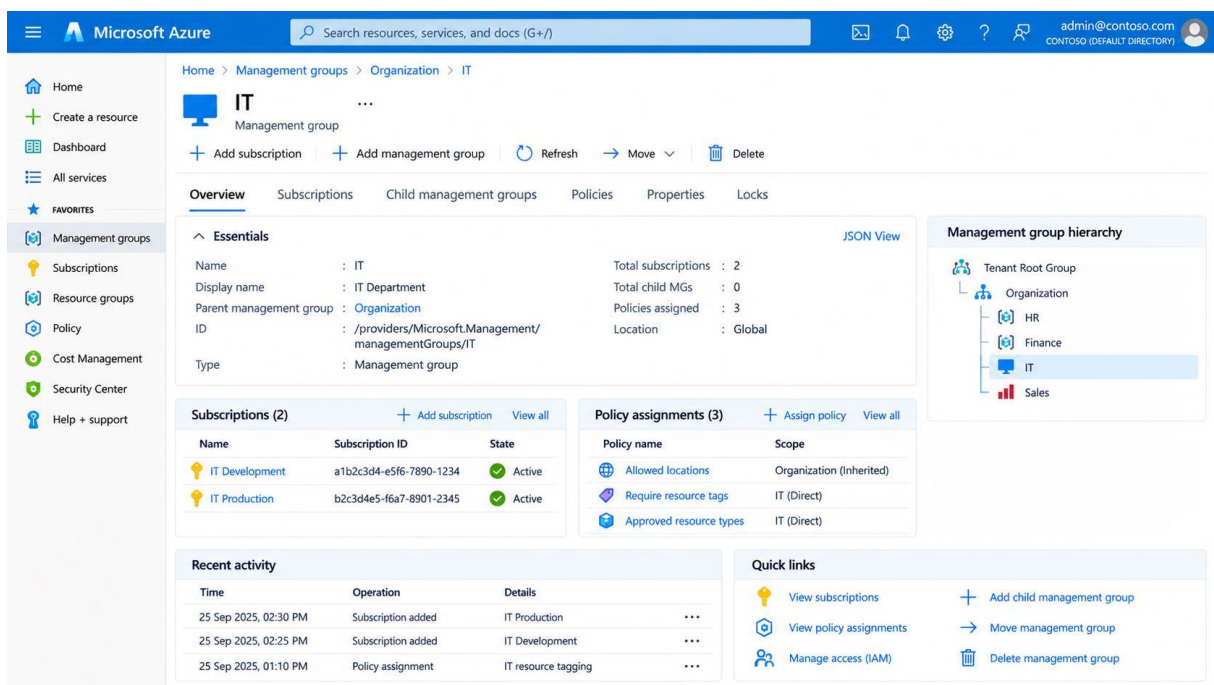


Figure 9: IT Management Group – Overview